

MOHAMED OMAR ATA

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer passionate about building intelligent systems on top of Large Language Models, with strong foundations in Data Science and end-to-end Machine Learning. Experienced in designing production-grade RAG pipelines, multi-agent architectures, and LLM-powered applications using LangChain, LangGraph, and FastAPI. Skilled at taking AI systems from idea to deployment by combining deep learning, classical ML, and cloud-native engineering.

PROFESSIONAL EXPERIENCE

Freelance AI Engineer

May 2026 (3-week engagement)

Client Project — Food Safety RAG System

- Designed and delivered a production-grade RAG system for food safety compliance, answering complex regulatory and risk queries with full citations and near-zero hallucination.
- Built hybrid retrieval (FAISS + BM25 with RRF fusion) and Cross-Encoder reranking, plus table-aware extraction and per-page semantic chunking.
- Achieved +84% Token F1 and +70% MRR over baseline through a reproducible evaluation benchmark.
- Deployed on AWS (ECS Fargate + ECR) with a Dockerized FastAPI backend.

Lead AI Instructor & Founder

2026 — Present

NeuroStack — Online AI Academy

- Founded an online AI academy delivering a full AI Engineering diploma that takes learners from Python fundamentals to production-grade AI systems.
- Designed the complete curriculum covering: Python & OOP, Data Handling (NumPy, Pandas), Data Visualization, Data Preprocessing, Machine Learning, Deep Learning, Computer Vision, NLP, LLM Fundamentals, RAG Systems, AI Agents, and MLOps.
- Positioned the program around practical, deployable AI projects that bridge theory with real-world engineering practice.

AI Engineering Instructor

2024 — Present

Quarter Academy & T-Square, Mansoura

- Delivered a comprehensive AI engineering curriculum to 40+ students per cohort, covering the full spectrum from Python fundamentals to advanced LLM applications.
- Taught a structured diploma covering: Python & OOP, NumPy, Pandas, Data Visualization, Preprocessing, Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, Prompt Engineering, LLM APIs, RAG Systems, and AI Agents (LangChain, LangGraph).
- Achieved high completion rates through project-based learning that bridges theory and real-world implementation.

Data Scientist Trainee

Dec 2024 — May 2025

Pioneers of Digital Egypt — AI & Data Science (IBM Program)

- Implemented end-to-end ML solutions including predictive modeling, data analysis, and MLOps practices using Python, Scikit-learn, and MLflow.
- Built and evaluated multiple ML models on real-world datasets as part of the IBM-backed DEPI national AI program.

AI Trainee

Aug 2025 — Sep 2025

NTI — Huawei AI Scholarship

- Specialized in advanced deep learning: computer vision, time series analysis, and generative models using TensorFlow, Keras, CNNs, RNNs, and GANs.
- Developed an AutoML graduation project covering end-to-end pipelines for structured data and computer vision tasks.

Django Full Stack Intern

Jul 2025 — Aug 2025

Information Technology Institute (ITI)

- Built and deployed full-stack web applications using Django, covering backend development and frontend integration.

PROJECTS

Multi-Agent RAG Assistant

github.com/M0hamed0mar/multi-agent-rag-assistant

Production-grade agentic RAG system built with LangGraph and Groq LLM.

- Engineered intelligent query routing across RAG, multi-step reasoning, tool usage, and chat workflows.
- Implemented dynamic tool-calling agents (Python, calculator, web search) with self-reflection and response validation.
- Built persistent SQLite memory and a modular FastAPI backend with real-time SSE streaming.
- Deployed on AWS (ECS Fargate + ECR + ALB) via Docker.

Tech Stack: Python, FastAPI, LangGraph, Groq LLM, ChromaDB, SQLite, Docker, AWS (ECS, ECR, ALB)

Adaptive Learning Agent

github.com/M0hamed0mar/adaptive-learning-agent

Multi-agent adaptive learning platform that generates personalized educational roadmaps from a free-form learning goal.

- Orchestrated a pipeline of 8 specialized LLM agents (Profile → Roadmap → Critic → Refiner → Teaching Brief → Tutor) with critique-and-improvement loops.
- Implemented lazy lesson generation, interactive tutor chat, and integrated web search for context-aware learning.
- Deployed end-to-end on AWS (ECS Fargate + ALB) via Docker.

Tech Stack: Python, FastAPI, LangGraph, LLM Agents, Web Search API, Docker, AWS (ECS, ECR, ALB)

Food Safety RAG System

github.com/M0hamed0mar/food-safety-rag

Domain-specific RAG for food safety compliance, delivered as a paid freelance engagement with measurable evaluation results.

- Built hybrid retrieval (FAISS + BM25 + RRF) with Cross-Encoder reranking, table-aware extraction (179 tables), and per-page semantic chunking.
- Achieved +84% Token F1 and +70% MRR over baseline through a reproducible evaluation benchmark.
- Engineered full citation support with near-zero hallucination.
- Deployed on AWS (ECS Fargate + ECR) with a Dockerized FastAPI backend.

Tech Stack: Python, LangChain, FAISS, BM25, Cross-Encoder Reranking, FastAPI, Docker, AWS (ECS, ECR)

Hybrid RAG Engine

github.com/M0hamed0mar/hybrid-rag-engine

Production-grade RAG engine for intelligent Q&A over multi-format user documents.

- Implemented hybrid search (FAISS + BM25) with RRF fusion, Cross-Encoder reranking, and multi-query expansion.
- Engineered intent-based query routing (factual / summary / reasoning / table) with response caching and persistent vector indices.
- Supported multi-format ingestion: PDF, DOCX, PPTX, images (OCR), and web pages.
- Deployed on AWS (ECS Fargate + ECR) via Docker.

Tech Stack: Python, LangChain, FAISS, BM25, OCR, FastAPI, Docker, AWS (ECS, ECR)

Rossmann Sales Forecasting Platform (Data Science)

github.com/M0hamed0mar/rossmann-sales-forecasting

End-to-end data science project forecasting daily sales for 1,115 drug stores across Germany.

- Engineered features with PCA and RobustScaler; trained XGBoost via RandomizedSearchCV achieving $R^2 \approx 0.97$ on the test set.
- Built interactive Plotly dashboards and a Flask web app supporting single-store and batch predictions.
- Deployed with a multi-stage Dockerfile and Docker Compose (Gunicorn) to AWS (ECS Fargate + ECR).

Tech Stack: Python, Scikit-learn, XGBoost, PCA, Plotly, Flask, MLflow, Docker, AWS (ECS, ECR)

EDUCATION

Bachelor of Science in Mechatronics Engineering

Expected Graduation: 2027

Faculty of Engineering, Tanta University

- Cumulative Grade: Very Good | Last Year Grade: Very Good

TECHNICAL SKILLS

LLM & Agent Engineering: LangChain, LangGraph, RAG Systems, AI Agents, Prompt Engineering, LLM APIs (OpenAI, Groq), Hugging Face, Fine-Tuning, Vector Databases (FAISS, ChromaDB), Semantic Search, Structured Outputs, Memory Systems, LLM Orchestration

NLP: Transformers, BERT, GPT, LLaMA, Mistral, Gemini, Text Classification, Sentiment Analysis, NER, Summarization, Question Answering

Deep Learning & Computer Vision: PyTorch, TensorFlow, Keras, CNNs, RNNs, LSTMs, GANs, Vision Transformers (ViT), Transfer Learning, Object Detection, Image Segmentation, ResNet, VGG

Machine Learning & Data Science: Scikit-learn, XGBoost, LightGBM, CatBoost, Random Forest, SVM, Ensemble Methods, Hyperparameter Tuning, PCA, K-Means, MLflow, Pandas, NumPy, Feature Engineering, EDA, Statistical Analysis, Matplotlib, Seaborn, Plotly, Dash

Cloud & DevOps: AWS (ECS Fargate, ECR, ALB, S3, Lambda, CloudWatch), Docker, Docker Compose, GitHub Actions, FastAPI, Flask, Gunicorn

Databases & Tools: ChromaDB, FAISS, PostgreSQL, MySQL, SQLite, Git, GitHub, Jupyter, VS Code, Google Colab

Backend Development: FastAPI, Flask, Django, RESTful APIs, asyncio, Regex

Mathematics & Statistics: Probability, Statistics, Linear Algebra, Calculus, Distributions

CERTIFICATIONS & COURSES

- Machine Learning Specialization — Andrew Ng, Coursera — [View](#)
- Deep Learning Specialization — Andrew Ng, Coursera — [View](#)
- AI Engineering Specialization — IBM, Coursera (RAG, AI Agents, LLMs) — [View](#)
- DEPI Program Certificate — Data Science & AI — [View](#)
- NTI Certificate — AI & Machine Learning — [View](#)
- Generative AI — Scrimba, Coursera — [View](#)
- IBM SkillsBuild Certificate — Artificial Intelligence Fundamentals — [View](#)
- AI and Data Science Diploma — Epsilon AI

LANGUAGES

Arabic: Native | **English:** B2 (Very Good) | **German:** B1